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Advance Digital Logic
EEDG/CE 6301
Spring 2026 Project v1.0

VERSION Evan Brock Wu

Instructions

This is your personalized project.

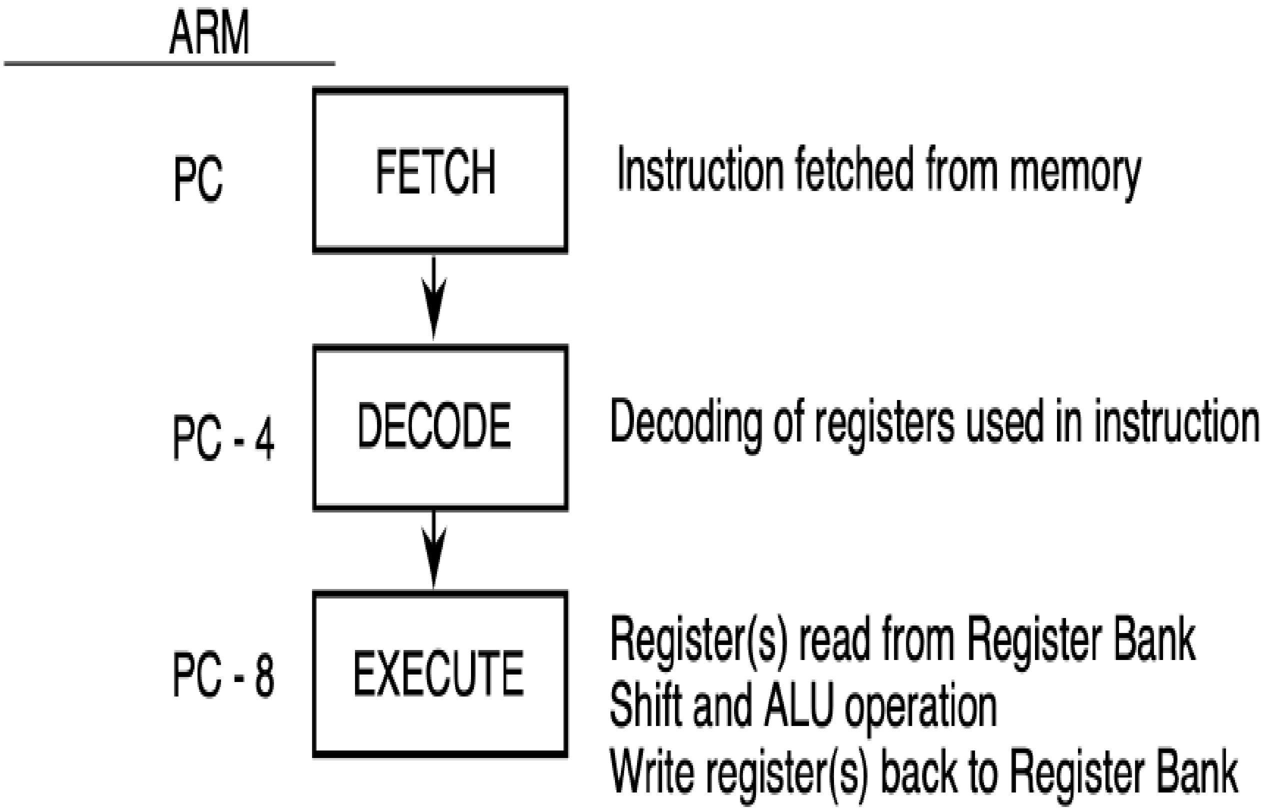
Consultation with any person (other than the TAs or Professor) at any location constitutes cheating. However, you are free to visit the Web and explore solutions to your design problem. In fact, ASIC World at http://asicworld.com and similar sites are encouraged.

If cheating is detected, the incident will be dealt with according to established UT-Dallas procedures. Possible disciplinary actions for academic dishonesty include a failing grade on this project.

Problem Statement—Version ebw220000

In this project, you are to implement an ARM 32 bit processor in System Verilog. This semester we will implement a processor for the ARM instruction set. Below in Figure 1 is block diagram of the ARM architecture. You are to implement a subset of the instruction set. The documentation for this device has been uploaded to eLearning. Many of these instructions have multiple variations (usually due to addressing). You may pick the variation of your choice.

Figure 1: ARM block diagram



In order to prevent AI generated answers, your design must use what we will call reverse endian, that is all arrays must be allocated as least significant bit to most significant bit. No credit will be given for any projects which don't follow this rule. For example, you must the following order:

```
logic mydata[0:63] ;
```

For more information, see https://en.wikipedia.org/wiki/ARM_architecture_family

You are to implement and test the following program as a minimum for a passing grade:

C := A SUB B

where A = 24 and B = 20.

In addition, you are to create a program that loops over this operation for 10 iterations, thereby accumulating a result:

```
C := 0 ;
for  i := 1 ; i <= 10 ; i = i + 1 do
     C := C ADD i

The grading will be as follows:
Submitted code:              +50
Code passes syntax check:    +10
Code is synthesizeable :     +10
Minimal program works:       +10
Looping program works:       +10
Random program works:        +10
---------------------------------
                             100
```

Specifications for ebw220000

Instructions to be Implemented
Arithmetic Commands: SUB ADD
Logical Commands: CMN TEQ
Data Transfer Commands: LDR STR
Boolean Commands: MVN AND
Instruction Control Commands: B BL
Memory Addressing Endianess : Little
Internal Array indices must be [LSB:MSB] or reverse endian

An example architecture can be show in this drawing below (Figure 2).

Figure 2: Example ARM design